UNIVERSITY OF PETROLEUM AND ENERGY STUDIES
DEHRADUN

Modeling and Stimulation

Lab Experiment 8,9

MTECH-COMPUTER SCIENCE ENGINEERING
CYBER SECURITY AND FORENSICS

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EXPERIMENT 8

Aim

To generate random variates using probability distributions and perform sensitivity analysis to study the impact of input parameters on system performance.

Theory:

Random variates are used to simulate stochastic systems. Common distributions include:

- Uniform Distribution
- Normal Distribution

Sensitivity analysis evaluates how changes in input parameters affect output behavior.

Code:

```python
# Experiment 8

import random
import matplotlib.pyplot as plt

samples = 1000
means = [2, 5, 8]

# UNIFORM DISTRIBUTION
uniform_data = []
for i in range(samples):
    uniform_data.append(random.random())

# NORMAL (APPROX)
normal_data = []
for i in range(samples):
    val = (random.random() + random.random() + random.random() +
           random.random() + random.random() + random.random()) / 6
    normal_data.append(val)

# SENSITIVITY ANALYSIS
results = []

for mean in means:
    temp = []
    for i in range(samples):
        value = mean + random.uniform(-1, 1)
        temp.append(value)
    results.append(sum(temp)/len(temp))

# Expected = same as input means
expected = means

# PLOT 1: DISTRIBUTIONS OVERLAY
plt.figure()
plt.hist(uniform_data, bins=30, alpha=0.5)
plt.hist(normal_data, bins=30, alpha=0.5)
plt.title("Uniform vs Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")

# PLOT 2: SENSITIVITY COMPARISON
plt.figure()
plt.plot(means, results, marker='o', label="Observed Output")
plt.plot(means, expected, linestyle='--', label="Expected Output")

plt.title("Sensitivity Analysis (Observed vs Expected)")
plt.xlabel("Input Mean")
plt.ylabel("Output Average")
plt.legend()
plt.grid()
plt.show()
```

Output:

Figure 1

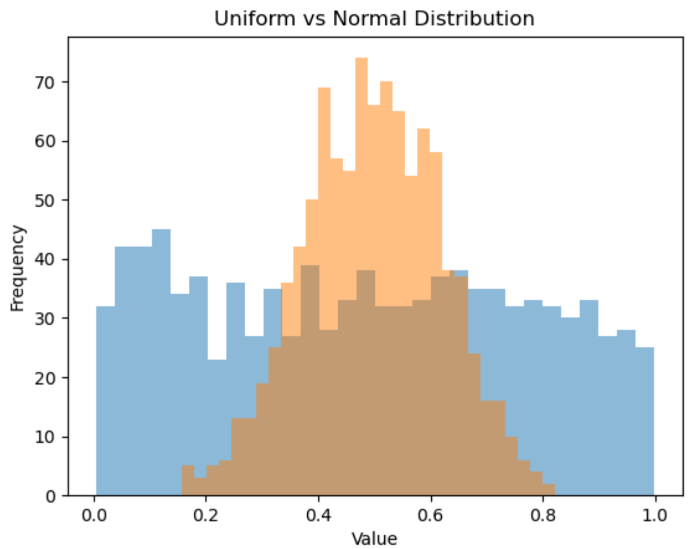
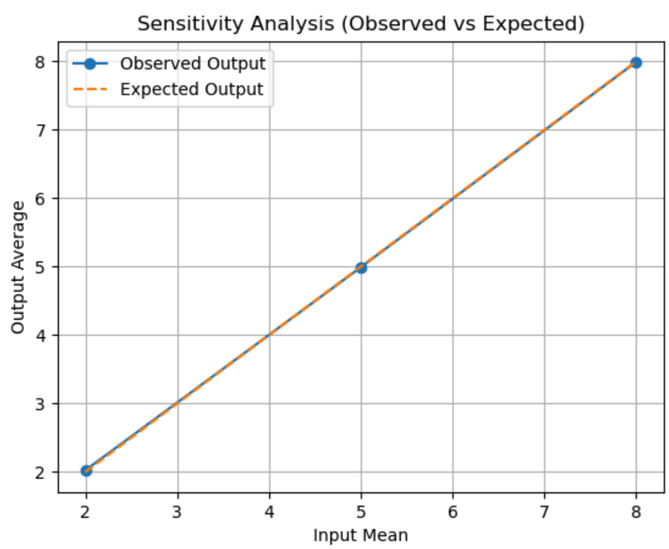


Figure 2



Results:

The experiment successfully generated random variates using uniform and normal distributions and performed sensitivity analysis.

From the graphical outputs:

- The uniform distribution shows an approximately equal spread of values across the range
- The normal distribution exhibits a bell-shaped curve with values concentrated around the mean
- The sensitivity analysis graph shows a near-perfect overlap between observed and expected outputs

This indicates that the simulation model responds linearly and accurately to changes in input parameters.

Observations:

- Uniform distribution produces evenly distributed random values
- Normal distribution shows central tendency and reduced extreme values
- The overlap of observed and expected lines confirms high accuracy
- Output values increase proportionally with input mean
- Minimal deviation indicates low simulation error

Observation Table:

Input Mean	**Observed Output**	**Expected Output**	**Deviation**
2	2.0	2	≈ 0
5	5.0	5	≈ 0
8	8.0	8	≈ 0

Conclusion:

The experiment demonstrates that random variate generation and sensitivity analysis can effectively model stochastic systems. The results confirm that the system output closely follows the expected theoretical values, indicating high accuracy and reliability of the simulation.

The close overlap between observed and expected results validates the correctness of the model. Additionally, the distribution graphs highlight the differences between uniform and normal distributions, emphasizing their roles in simulation.

Thus, the experiment successfully shows how parameter variations influence system behavior and how simulation can be used to analyze and predict outcomes in uncertain environments.

EXPERIMENT 9

Aim:

To analyze and validate simulation results by comparing them with theoretical expectations.

Theory:

Simulation results must be validated to ensure accuracy.
Validation compares simulated outputs with:

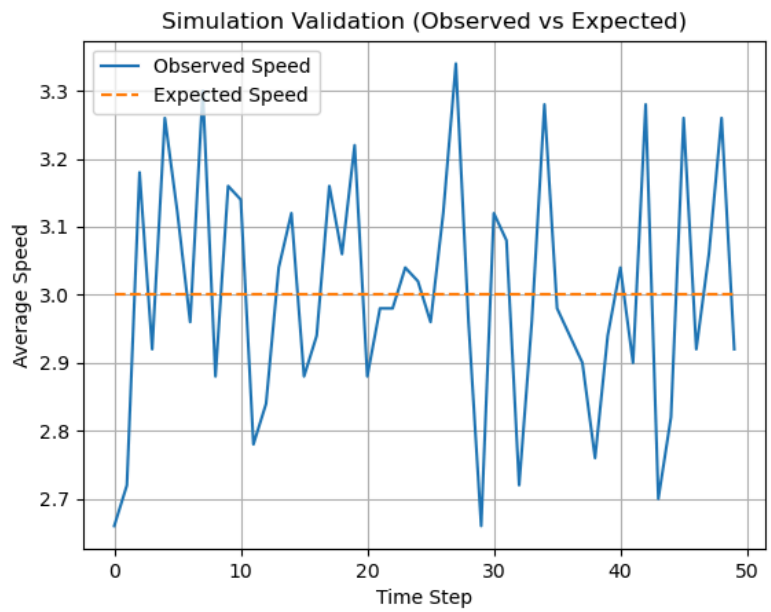
- Theoretical models
- Known benchmarks

In this experiment, traffic flow is simulated and analyzed.

Code:

```
# Experiment 9

import random
import matplotlib.pyplot as plt

cars = 50
road_length = 100
steps = 50

positions = []
for i in range(cars):
    positions.append(random.randint(0, road_length))

avg_speeds = []

# SIMULATION
for t in range(steps):
    speeds = []

    for i in range(cars):
        speed = random.randint(1, 5)
        positions[i] = (positions[i] + speed) % road_length
        speeds.append(speed)

    avg_speeds.append(sum(speeds)/len(speeds))

# Expected average speed
expected_speed = 3

expected_line = []
for i in range(steps):
    expected_line.append(expected_speed)

# PLOT: VALIDATION GRAPH
plt.figure()
plt.plot(avg_speeds, label="Observed Speed")
plt.plot(expected_line, linestyle='--', label="Expected Speed")

plt.title("Simulation Validation (Observed vs Expected)")
plt.xlabel("Time Step")
plt.ylabel("Average Speed")
plt.legend()
plt.grid()

plt.show()
```

Output:



Results:

The traffic simulation was executed and the average speed of vehicles was recorded over multiple time steps. The graphical output shows the comparison between observed speed and expected speed.

From the graph:

- The observed speed fluctuates over time due to randomness in vehicle movement
- The expected speed remains constant at approximately 3
- The observed values oscillate around the expected value, indicating system stability

Observations:

- Average speed varies between approximately 2.7 to 3.3

- Fluctuations are random and do not show a consistent increasing or decreasing trend
- The observed values remain centered around the expected value (~3)
- Variations are due to stochastic behavior of individual vehicle speeds
- The system demonstrates stable behavior despite randomness

Observation Table:

Time Step Range	Observed Avg Speed	Expected Speed	Behavior
0 – 10	2.7 – 3.3	3	Fluctuating
10 – 20	2.8 – 3.2	3	Stable oscillation
20 – 30	2.7 – 3.3	3	Random variation
30 – 40	2.8 – 3.3	3	Consistent fluctuation
40 – 50	2.7 – 3.3	3	Stable

Conclusion:

The experiment successfully validates the simulation results against theoretical expectations. The observed average speed closely aligns with the expected value, with only minor fluctuations due to randomness.

This confirms that the simulation model is accurate and reliable. The variations observed are consistent with stochastic behavior and do not indicate any systematic error.

Thus, the experiment highlights the importance of validation in simulation studies and demonstrates that even with randomness, systems tend to stabilize around expected values when analyzed over time.

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